

Math 107-01
Test 1A: September 16, 2013

Answer all of the following questions. You will have 50 minutes to complete the test. You may not use any outside assistance on this test. You may use a calculator, but the use of electronic equipment such as mp3 players, ipods, cell phones and any electronic devices other than your calculator during the test is prohibited. If you have time, check your answers for accuracy.

Be sure to show all work. Unsupported answers may not receive credit.

Good luck!

Name: _____

1. Let $U = \{a, b, c, d, e, f, g, h, i, j, k\}$ be the universal set for this problem. Let $A = \{b, d, e, f, g, h\}$, $B = \{a, c, e, h, j\}$ and $C = \{c, i, j, k\}$ be subsets of U . Find the following (2 points each):

(a) $A \cap B$

(b) $A \cup B$

(c) A'

(d) B'

(e) C'

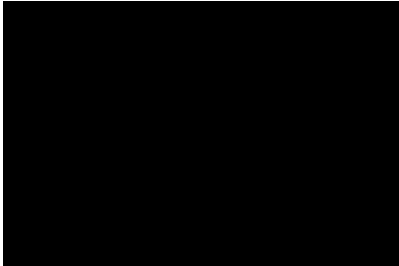
(f) $A \cap B'$

(g) $B \cup C'$

(h) $A' \cup C$

(i) $(B \cap C)'$

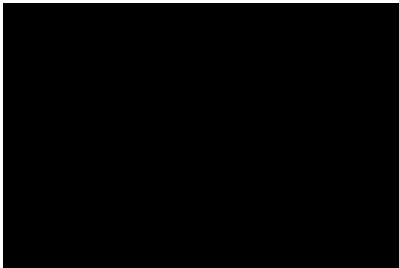
2. A survey of 1329 high school students revealed the following: 719 attended football games, and 878 attended basketball games. 300 of those who attended football games did not attend basketball games. Let F be the set of those who attended football games and B the set of those who attended basketball games. Fill in the following Venn diagram to illustrate the results of this survey (8 points):



3. A group of 1500 high school seniors were surveyed, and the following information resulted: 774 have read The Hunchback of Notre Dame, 738 have read The Scarlet Letter, 819 have read The Once and Future King, 200 have read all three books, 386 have read The Hunchback of Notre Dame and The Scarlet Letter, 389 have read The Scarlet Letter and The Once and Future King, 170 have read only The Hunchback of Notre Dame

Fill in the appropriate numbers to make a Venn diagram illustrating these results.

Let HN represent the set of those who had read The Hunchback of Notre Dame, SL the set of those who had read The Scarlet Letter, and OF the set of those who had read The Once and Future King. (8 points)



4. Let $n(U) = 125$. Also, let A and B be subsets of U with $n(A) = 15$ and $n(B) = 50$. (3 points each)

(a) Find $n(A')$.

(b) If $n(A \cap B) = 8$, find $n(A \cup B)$.

5. A group of 200 MHC students was surveyed about the type of vehicle that they drive. The survey collected the following data:



where B represents the set of those who drive a blue vehicle and T represents the set of those who drive a truck. Answer each question according to the Venn diagram provided (2 points each):

(a) How many students drive a blue vehicle that is not a truck?

(b) How many students drive a blue truck?

(c) How many students drive truck that is not blue?

6. A group of 400 MHC students was surveyed about their family pets: whether their families owned a cat, a dog, or fish. The survey collected the following data:

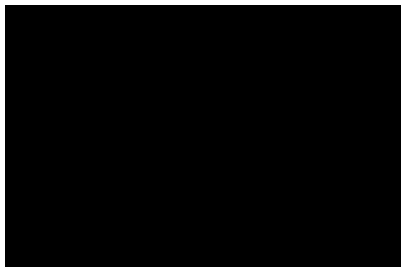


where C represents the set of those who own a cat, D represents the set of those who own a dog, and F represents the set of those who own fish. Answer each question according to the Venn diagram provided (2 points each):

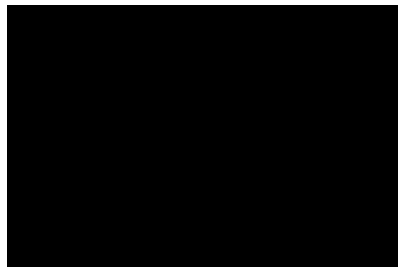
- (a) How many families have only a cat?
- (b) How many families have both a cat and a dog?
- (c) How many families have a cat or a dog?
- (d) How many families have all three types of pet?
- (e) How many families have none of these pets?
- (f) How many families have both a cat and fish?
- (g) How many families have a cat or fish?

7. (2 points) Which of the following Venn diagrams represents $A \cap B'$? Circle your answer.

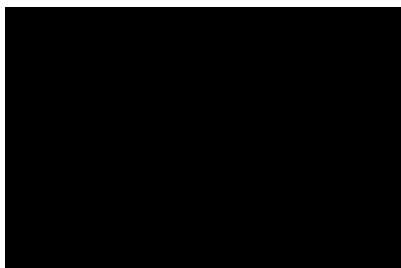
A



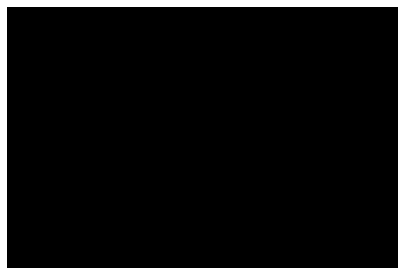
B



C



D



8. Evaluate each of the following (2 points each):

(a) $5!$

(b) $0!$

(c) $4!$

(d) $\frac{17!}{14!}$

(e) ${}_6P_2$

(f) ${}_{11}C_5$

$$(g) \quad \frac{6!}{3!}$$

9. The Create-a-Cat Store allows customers to create their own personal cat stuffed animal. Customers must choose the following: size (small, medium, large), color of fur (6 options), and color of kitty collar (5 options). How many different cat stuffed animals can be created? (4 points)
10. Suppose six teams enter a mathematics tournament. How many different ways can the teams rank at the end of the tournament? (5 points)
11. Suppose a school issues parking tags with identification numbers consisting of three digits (which may be repeated) followed by one letter from the following: A,B,C,D. How many different identification numbers are possible? (5 points)
12. The local branch of a national business chain has 27 employees. They are required to send 5 employees to a training session at the national headquarters. How many ways can they select the five employees to send? (5 points)

13. A certain pet store has 26 fish and 15 lizards available for purchase. How many ways can a customer purchase 5 fish and 4 lizards? (5 points)

On my honor, I have neither given nor received any academic aid or information that would violate the Honor Code of Mars Hill University. _____(please sign)